

MA 221: Numerical Analysis

Course Code: MA 221

Course Name: Numerical Analysis

L-T-P-C: 2-0-3-4

Prerequisites: Computer Organization and Processor Architecture Design (EE 326) or equivalent.

Students intended for: B. Tech. 2nd year students

Elective or Core: Core for B. Tech. Mathematics and Computing, Elective for other disciplines.

Mutual Exclusion: MA 523

Approval: 59th BoA

Course Contents

- **Introduction to Numerical Computing:** Fixed-point Representation, Floating-point Representation, Significant digits, Approximations and Errors in Numerical Computing, Error Propagation, Conditioning & Stability. [4 Lectures]
- **Roots Finding Methods for Nonlinear Equations:** Introduction, Bisection Method, Method of False Position, Secant method, Fixed-point iteration method, Newton-Raphson Method; Roots Finding Methods for a system of Non-linear equations; Roots Finding Methods for a Complex equation. [6 Lectures]
- **Solution of the System of Linear Equations:** Direct Solution process: Matrix Inversion Method, Gauss Elimination Method, Partial-Pivoting, Gauss-Jordan Method; Iterative Solution process: Jacobi Iteration Method, Gauss-Seidel Method, Method of Relaxation, QR decomposition Method, Convergence of Iteration Methods. [5 Lectures]
- **Curve Fitting using numerical methods:** Interpolating polynomials, Lagrange Interpolation Polynomial; Newton Interpolation Polynomial: Newton Divided Difference formula, Newton Forward Difference formula, Newton Backward Difference formula; Cubic spline, Regression Curves, Least-Square approximations. [6 Lectures]
- **Numerical Differentiation:** Forward, Backward and Central Difference, Richardson's Extrapolation formula. [3 Lectures]
- **Numerical Integration:** Newton-Cotes formula, Trapezoidal Rule, Simpson's One-Third Rule, Simpson's Three-Eight formula, Romberg Integration, Gaussian quadrature. [4 Lectures]

Laboratory/practical/tutorial Modules:

The computer lab classes will complement the theoretical understanding of the students. All the numerical methods, discussed in the lecture classes, will be used to solve mathematical/physical problems through computer programming during the lab classes. This will provide a hands-on experience to the students on this subject.

Text books:

1. K. E. Atkinson, **An Introduction to Numerical Analysis**, 2nd Edition, John Wiley, 2008.
2. R. L. Burden and J. D. Faires, **Numerical Analysis**, 7th Edition, Thomson Learning, 2001.



References:

1. M. T. Heath, **Scientific Computing: An Introductory Survey**, McGraw Hill, 2002.
2. Brian Bradie, **A friendly introduction to Numerical Analysis**, Pearson Education, 2007.